(see 4.8)
A-11 Appropriate measures should be provided to prevent vents becoming blocked by accumulations of ice formed from moisture in the air.
Low temperature
(see 4.2.2)
A-12 Due consideration should be given to means for handling boil-off gas.
High pressure
(see 4.8)
A-13 Due consideration should be given to static electricity associated with rotating or reciprocating machinery including the installation of conductive machinery belts and precautionary measures incorporated in operating and maintenance procedures , in addition to the bonding of tanks, piping and equipment required by paragraph 5.7.4 of the Code.

SOURCE: MSC.562(108).pdf
URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.562(108).pdf
addition to the bonding of tanks, piping and equipment required by paragraph 5.7.4 of the Code.
Anti-static clothing and footwear, and a portable hydrogen detector should be provided for each crew member working in the cargo area.
Static electricity
(see 4.9.2)
A-14 The cargo operation manuals required in paragraph 18.2 of the Code should include limitations of various operations in relation to environmental conditions.
Wide range of
flammability limits
(see 4.10)
RESOLUTION MSC.565(108) (adopted on 24 May 2024)
REVISED INTERIM RECOMMENDATIONS FOR CARRIAGE OF LIQUEFIED HYDROGEN IN BULK

No. Special requirement Related hazard
A-15 An appropriate procedure should be established for warm -up, inert gas purge, gas -free, hydrogen purge and pre -cooling.
The procedure should include:
.1 selection of inert gas in relation to temperature limit;
.2 measurement of gas concentration;

SOURCE: MSC.562(108).pdf
URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.562(108).pdf
The procedure should include:
.1 selection of inert gas in relation to temperature limit;
.2 measurement of gas concentration;

.3 measurement of temperature;
.4 rates of supply of gases;
.5 conditions for commencement, suspension, resuming and termination of each operation;
.6 treatment of return gases; and
.7 discharge of gases.

Prevention of dangerous purging operation (see 4.11)

A-16 Only almost pure para-hydrogen (i.e. more than 95%) should be loaded in order to avoid excessive heating by ortho - to para-hydrogen conversion.

General (see 4.1)

A-17 Fire detectors for detecting hydrogen fire should be selected, taking into account the features of hydrogen fire, to the satisfaction of the Administration.

Features of hydrogen fire and fire hazard (see 4.7.4)

A-18 At the design stage, dispersion of hydrogen from vent outlets should be analysed in order to minimize risk of ingress of flammable gas into accommodation spaces, service spaces,

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should be analysed in order to minimize risk of ingress of flammable gas into accommodation spaces, service spaces,
machinery spaces and control stations. Extension of hazardous areas should be considered based on the results of the analysis.

Low density and high diffusivity (see 4.5)

A-19 Due consideration should be given to appropriate safety measures to prevent formation of explosive mixture in the case of a leakage and permeation of hydrogen, including:
.1 installation of hydrogen detectors in order to detect a possible ground -level travel of low temperature hydrogen gas, and at high points in spaces where warm hydrogen gas can be trapped; and
.2 application of "best practice" for land -based liquid hydrogen storage taking into account appropriate guidance such as "Cryogenics Safety Manual Fourth Edition (1998)"8).

General (see 4.1)

A-20 In the case that fusible elements are used as a means of fire

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SOURCE: MSC.562(108).pdf
URL:

https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.562(108).pdf
Edition (1998)"8).
General
(see 4.1)
A-20 In the case that fusible elements are used as a means of fire detection required by paragraph 18.10.3.2 of the Code, flame detectors suitable for hydrogen flames should be provided in addition at the same locations. Appropriate means should be adopted to prevent the activation of ESD system owing to false alarm of flame detectors, e.g. avoiding activation of ESD system by single sensor (voting method).
Fire hazard
(see 4.7.4)
RESOLUTION MSC.565(108) (adopted on 24 May 2024)
REVISED INTERIM RECOMMENDATIONS FOR CARRIAGE OF LIQUEFIED HYDROGEN IN BULK
MSC 108/20/Add.1
Annex 20, page 7
I:\MSC\108\MSC 108-20-Add.1.docx
No. Special requirement Related hazard
A-21 Consideration should be given to enhance the ventilation capacity of the enclosed spaces subject to liquefied hydrogen leakage and permeation, taking into account the latent heat of vaporization, specific heat and the volume of hydrogen gas in
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SOURCE: MSC.562(108).pdf
URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.562(108).pdf
leakage and permeation, taking into account the latent heat of vaporization, specific heat and the volume of hydrogen gas in relation to temperature and heat capacity of adjacent spaces.
Low density and high diffusivity
(see 4.5)
A-22 Liquid and gas hydrogen pipes should not pass through enclosed spaces in addition to other than those referred to in paragraph 5.2.2.1.2 of the Code, unless:
.1.1 the spaces are equipped with gas detection systems which activate the alarm at not more than 20% LFL and shut down the isolation valves, as appropriate, at not more than 40% LFL (see sections 16.4.2 and 16.4.8 of the Code); and
.1.2 the spaces are adequately ventilated; or
.2 the spaces are maintained in an inert condition.
Susceptibility to leakage (see 4.4)
A-23 A risk assessment should be conducted to ensure that risks arising from liquefied hydrogen cargo affecting persons on board, the environment, the structural strength or the integrity of the ship
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SOURCE: MSC.562(108).pdf
URL:

https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.562(108).pdf
arising from liquefied hydrogen cargo affecting persons on board, the environment, the structural strength or the integrity of the ship are addressed. Consideration should be given to the hazards associated with properties of liquefied hydrogen and hydrogen gas, physical layout, operation and maintenance, following any reasonably foreseeable failure. For the risk assessment, appropriate methods, e.g. HAZID, HAZOP, FMEA/FMECA, what-if analysis, etc., should be adopted taking into account IEC/ISO 31010:2019 "Risk management Risk assessment techniques"7) and SAE ARP 5580 -2001 "Recommended failure modes and effects analysis (FMEA) practices for non-automobile applications"9).

General

(see 4.1)

A-24 Relief valve sizing should be undertaken for the most onerous scenario. The evaluation should include the fire scenario and should consider the resulting magnitude of the heat flux on the cargo containment system.

High pressure

hazard

(see 4.8)

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should consider the resulting magnitude of the heat flux on the cargo containment system.

High pressure

hazard

(see 4.8)

A-25 A filling limit exceeding 98% at reference temperature should not be permitted.

High pressure

hazard